Probability (1): Midterm Exam

2019. 4. 22

Problem 1. *(10pts) Suppose n and m are positive integers. Suppose $\{X_1, X_2, \ldots, X_{n+m}\}$ are independent random variables. Show that the following two random vectors are independent:*

$$(X_1, \ldots, X_n), \quad (X_{n+1}, \ldots, X_{n+m}).$$

(1.1) (2pts) Write down the definition that $\{X_1, X_2, \ldots, X_{n+m}\}$ are independent.

(1.2) (2pts) Write down the definition that $(X_1, \ldots, X_n)$ and $(X_{n+1}, \ldots, X_{n+m})$ are independent.

(1.3) (6pts) Show that $(X_1, \ldots, X_n)$ and $(X_{n+1}, \ldots, X_{n+m})$ are independent.

Problem 2. *(40pts)*

(2.1) (15pts) Write down Borel-Cantelli Lemma and prove it.

(2.2) (5pts) Write down the definition of "uniform integrable" (UI).

(2.3) (10pts) Suppose X_n, X are integrable and $X_n \to X$ a.s. Show that

$$X_n \to X \text{ in } L^1 \quad \text{if and only if} \quad \{X_n, n \geqslant 1\} \text{ is UI.}$$

(2.4) (10pts) Let $\{X_n\}$ be i.i.d. and suppose $\mathbb{E}[|X_1|] = \infty$. Define $S_n = \sum_{j=1}^n X_j$. Show that almost surely $\limsup_{n\to\infty} |S_n|/n = \infty$.

Problem 3. *(10pts) Let $\{X_n\}$ be i.i.d. and the common law is* Exp(1)*: $\mathbb{P}[X_1 \geqslant x] = e^{-x}$ for $x \geqslant 0$. Show that*

$$\mathbb{P}\left[\limsup_{n\to\infty} \frac{X_n}{\log n} = 1\right] = 1.$$

Problem 4. *(20pts) For $X, Y \in L^2$, they are uncorrelated if* $\mathrm{cov}(X,Y) := \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$. *Suppose $\{X_n\}$ is a sequence of uncorrelated random variables. Suppose that there exists $M > 0$ such that* $\mathrm{var}(X_n) \leqslant M$ *for all n. Define $S_n = \sum_{j=1}^n X_j$. Show that*

$$\frac{S_n - \mathbb{E}[S_n]}{n} \to 0, \quad \textit{almost surely.}$$

(4.1) (5pts) Justify that it suffices to show the conclusion when $\mathbb{E}[X_n] = 0$ and $\mathbb{E}[X_n^2] \leqslant 1$ for all n; and we will assume such assumption in the following of the problem.

(4.2) (5pts) Show that $\mathbb{E}[S_n^2] \leqslant n$ and derive that

$$\lim_{n\to\infty} \frac{S_{n^2}}{n^2} = 0, \quad \textit{almost surely.}$$

(4.3) (5pts) For $n \geqslant 1$, define $\hat{n} = \lfloor \sqrt{n} \rfloor^2$. Set

$$Z_n := \frac{S_n - S_{\hat{n}}}{\hat{n}}.$$

Show that $\mathbb{E}[Z_n^2] \leqslant O(n^{-3/2})$.

(4.4) (5pts) Show the conclusion.

Problem 5. *(20pts) Suppose $\{X_n\}$ are i.i.d. Gaussian random variables with zero mean and unit variance. Define $S_n = \sum_{j=1}^n X_j$. Show that*

$$-1 \leqslant \liminf_n \frac{S_n}{\sqrt{2n\log\log n}} \leqslant \limsup_n \frac{S_n}{\sqrt{2n\log\log n}} \leqslant 1, \quad \textit{almost surely.}$$

(5.1) (3pts) For $\theta \in \mathbb{R}$, calculate $\mathbb{E}[\exp(\theta S_n)]$.

(5.2) (7pts) Show that, for any $c > 0$,

$$\mathbb{P}\left[\max_{k\leqslant n} S_k \geqslant c\right] \leqslant \exp\left(-\frac{c^2}{2n}\right).$$

(5.3) (3pts) Set $h(n) = \sqrt{2n\log\log n}$. For any fixed $\rho > 1$ and $N > 1$, show that

$$\sum_m \mathbb{P}\left[\max_{k\leqslant N^m} S_k \geqslant \rho h(N^m)\right] < \infty.$$

(5.4) (7pts) Show the conclusion.

Name: Student's ID:

1(10pts)	(2.1)(15pts)	(2.2-2.3)(15pts)	(2.4)(10pts)	3(10pts)	4(20pts)	5(20pts)	Total